Pixel: Experience from Run 3

Matej Roguljić, Johns Hopkins University

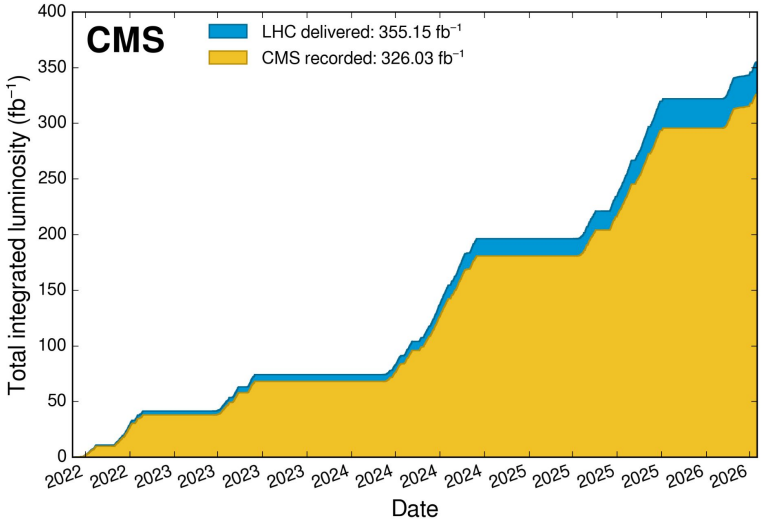
14.10.2026

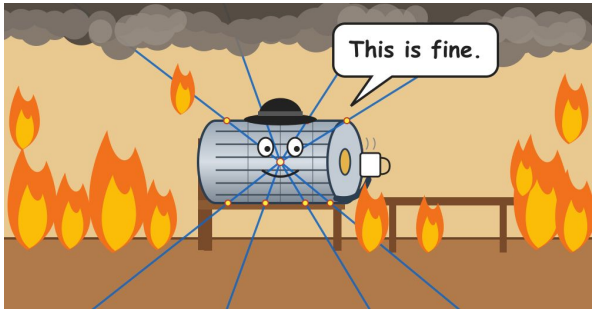
Introduction

- Sharing the lovely experience of monitoring, calibrating, and triaging the pixel detector in Run 3
- We learn most from problems so the talk focuses on specific issues we encountered
  - Covering several issues over the years so that we can learn for Ph2
  - Pixel detector kept providing good data even though we exceeded the design radiation tolerance of the innermost layer

With the innermost layer placed at a radius of 29 mm from the beam, the modules in this region have to withstand very high radiation doses and hit rates, as shown in Tab. 2. A hadron fluence of $3.6 \times 10^{15} n_{eq}/cm^2$ (fluence measured in units of 1 MeV neutron equivalents) is expected to be accumulated in the innermost layer after collecting an integrated luminosity of 500 fb^{-1}. This fluence is about twice as high as the operational limits of the proposed system, as defined by the charge collection efficiency of the sensor [4]. Therefore, the innermost BPIX layer will be replaced during LS2. The fluence in the second layer of the BPIX detector is about four times less, and hence the outer BPIX layers will stay operational during the entire period. The same is true for the modules in the FPIX

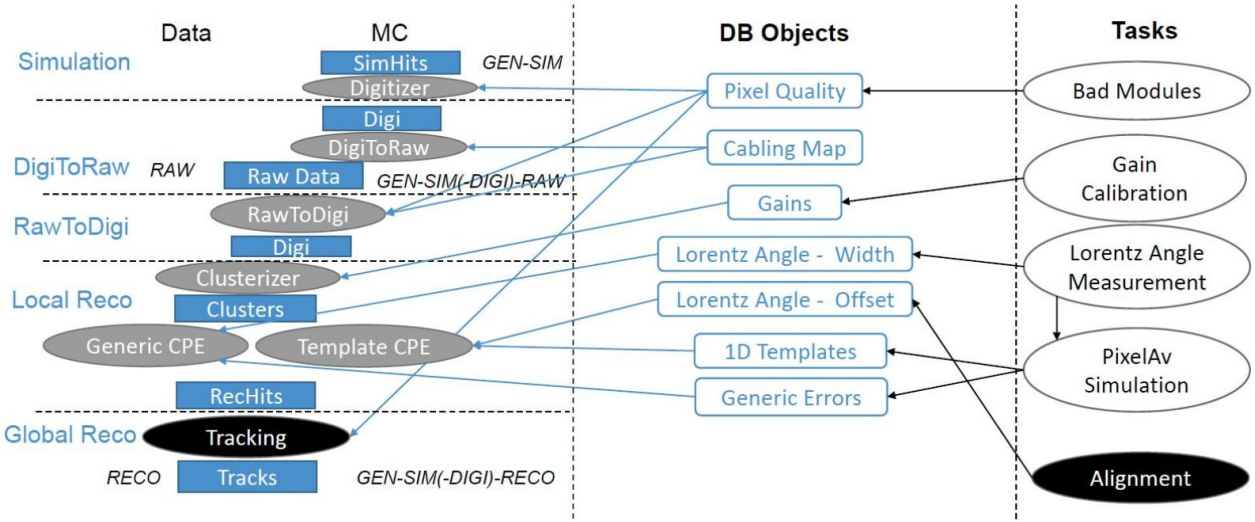
Phase 1 detector upgrade





Pixel local reconstruction responsibilities

- Detector calibrations
- Developing and maintaining CMSSW packages for pixel local reconstruction
  - Unpacking raw data, clustering hits, estimating cluster positions, etc.
- Performance measurements and monitoring
  - Performance notes found on this Twiki
- Detector simulation



Condition	For data or MC?
SiPixelFedCablingMap	Both
SiPixelGainCalibration	Both
SiPixelQualityFromDB	Both
SiPixelLorentzAngle SiPixelGenErrorDBObject SiPixelTemplateDBObject SiPixel2DTemplateDBObject	Both
SiPixelLorentzAngleSim SiPixel2DTemplateDBObject (denominator)	MC
SiPixelDynamicInefficiency	MC
SiPixelStatusScenarios SiPixelStatusScenarioProbability	MC
SiPixelVCal	Data (outside GT)

From Dinko's contribution at the Conditions preparations workshop (Indico) and Tracker Training Days (Indico)

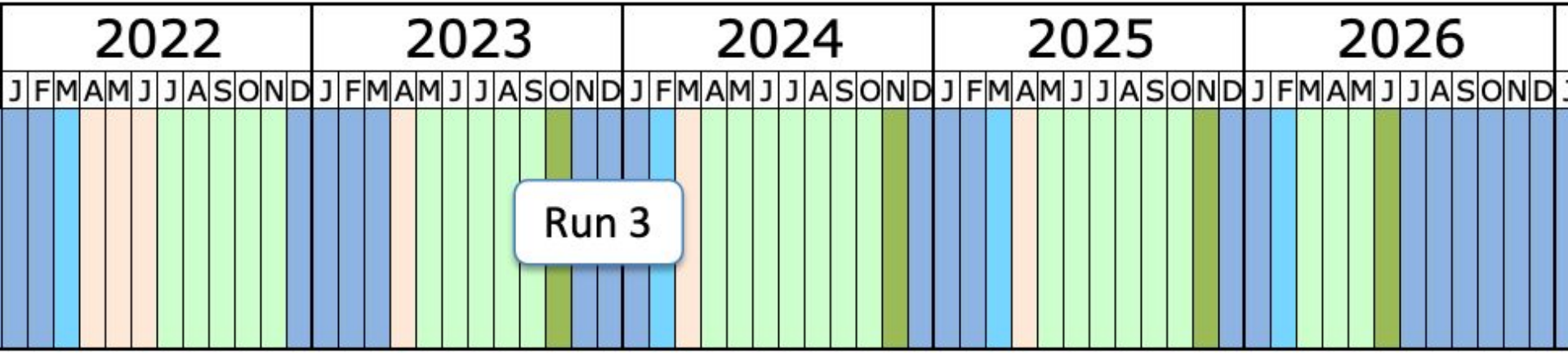
Timeline at a glance

Rapid sensor change at beginning of Run 3

L1 automasking becoming an issue in 2024

Hole in FPIX D-1

Multiple adjustments of reconstruction algorithm(s) to cope with radiation damage throughout 2025 and 2026



Overlapping dead modules in BPIX L3 / L4 in mid-2023

Late 2024 / early 2025 calibrations are unsuccessful: charge loss and cluster breakage problematic

Half of FPIX unresponsive at the very end of data-taking

Automasking up to ~30% in L1 in 2025

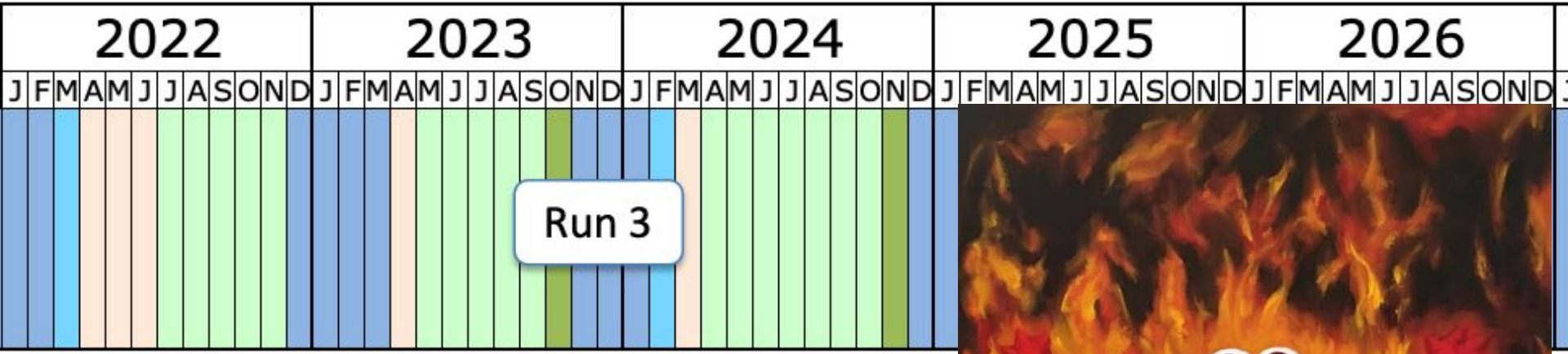
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Hole in FPIX D-1

Excellent period for practicing stress management



Overlapping dead modules in BPIX L3 / L4 in mid-2023

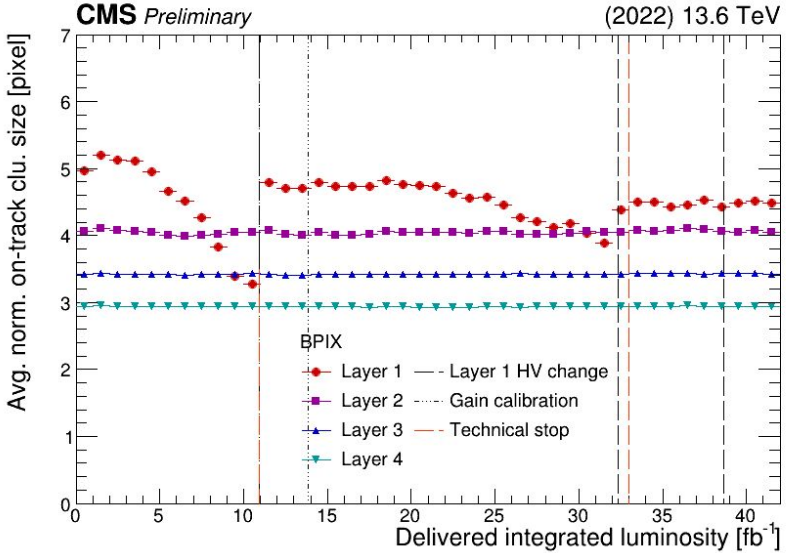
Late 2024 / calibrations are charge loss breakage p

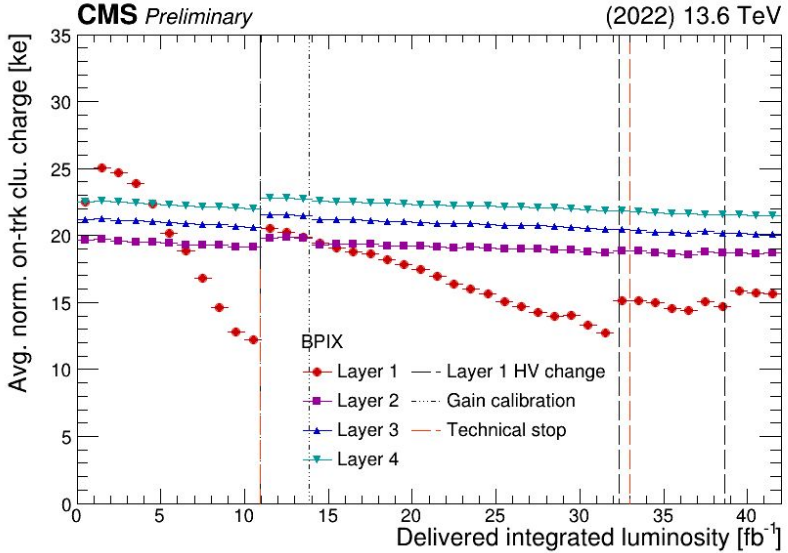


L1 in 2025

Rapid sensor change at beginning of Run 3

- New BPIX L1 sensor quickly changed with first runs
- The procedure for changing bias voltage and updating conditions did not allow for fast turnaround
  - Take one run at increased HV → **analyze data and produce templates** → derive appropriate alignment → full track validation → deploy
  - "Saved" by the technical stop where we had time without beam to derive and validate conditions for HV increase 150 → 300 V
- Lesson for Ph2: anticipate how fast the depletion voltage changes and revise procedure for increasing bias voltage

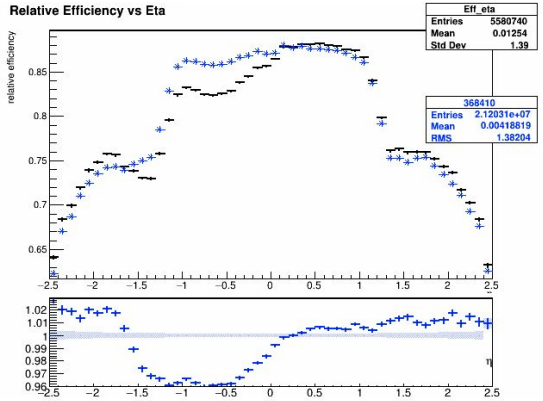


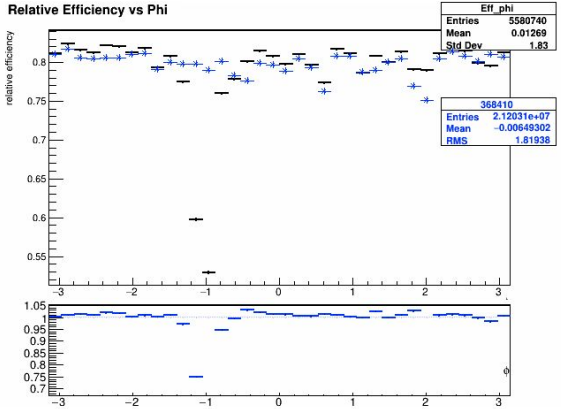


[2022 perf. plots]

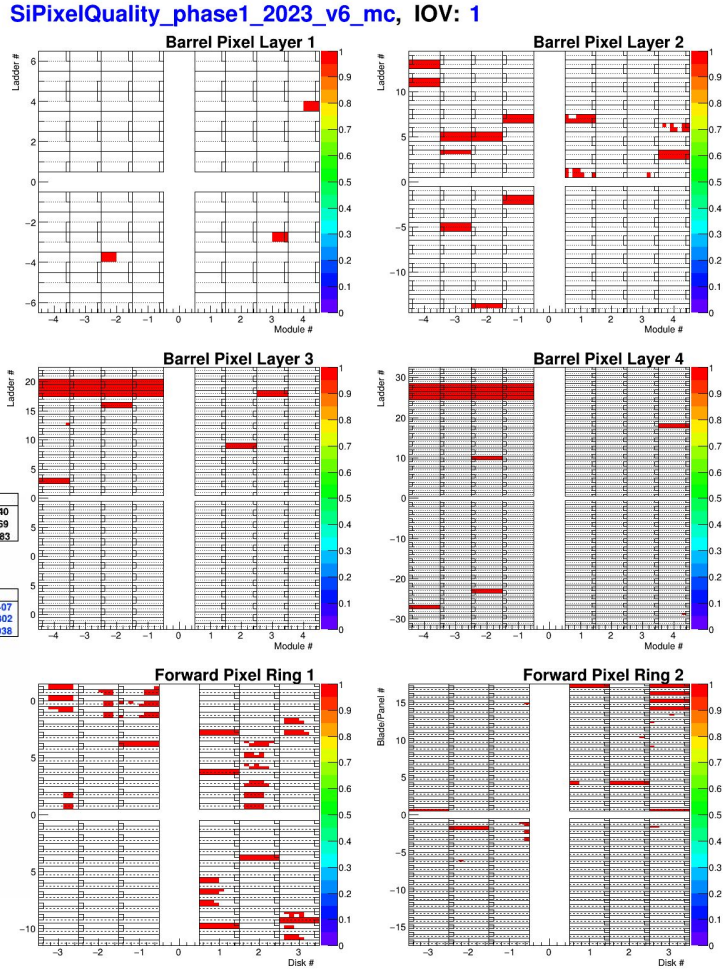
Overlapping holes in BPIX L3 and L4

- The bad modules are marked by PCL for prompt
  - Ph1 criterion is based on comparing ROC occupancies to average occupancies across the layer / disk
  - Required an atypical manual emergency intervention because it affected tracking at HLT
- Detection and flagging of problematic modules to be extended to HLT for Ph2
- Note: dominant effect in Run 3 was the infamous “automasking”, thankfully covered event-by-event with FedErrors



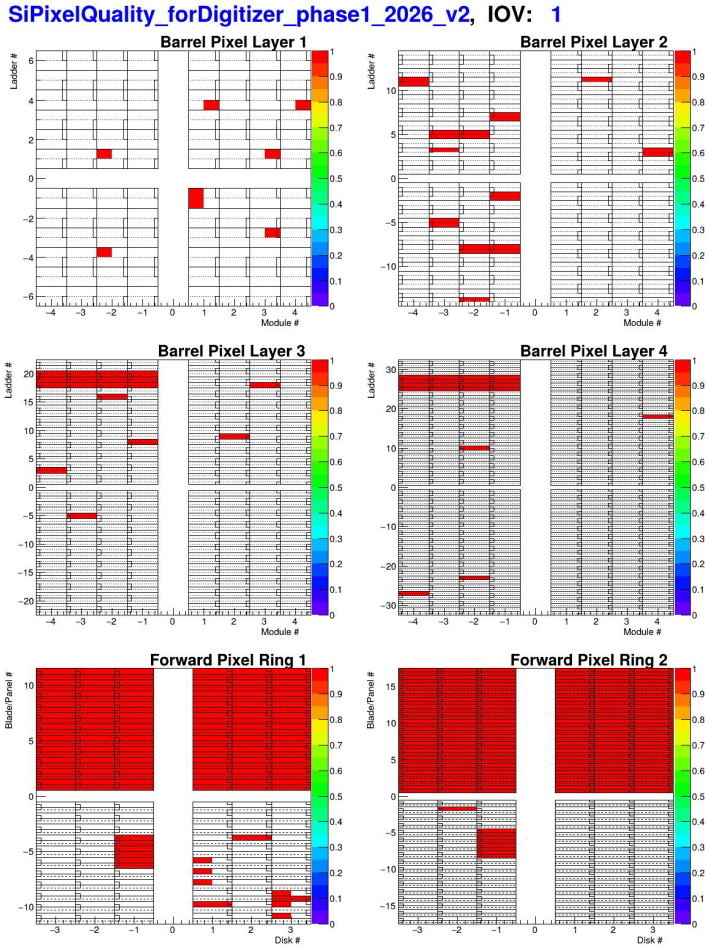


DQM link1, link2



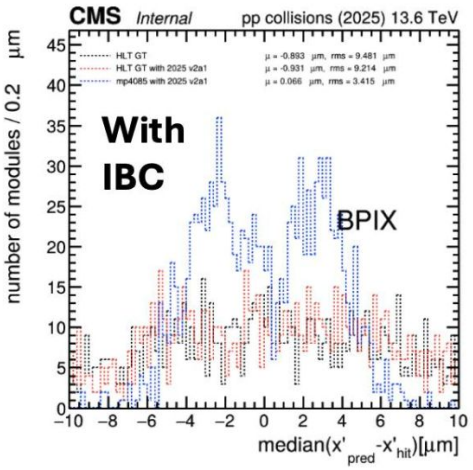
FPIX issues at the end of 2026

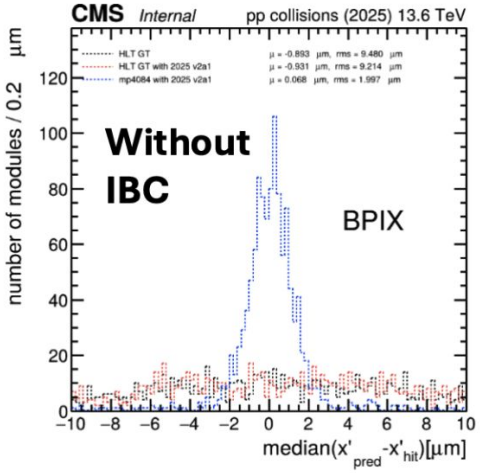
- At the very end of 2026 pp data-taking, half of FPIX became unconfigurable
- It was not clear if the problem was recoverable or not
  - Needed to create multiple payloads with some urgency to study the effects on HLT
    - What if we declare the modules as bad, but they do get recovered?
    - What if the modules are considered good, but they do not send out data?
  - Another example where automated calibrations for HLT are desired

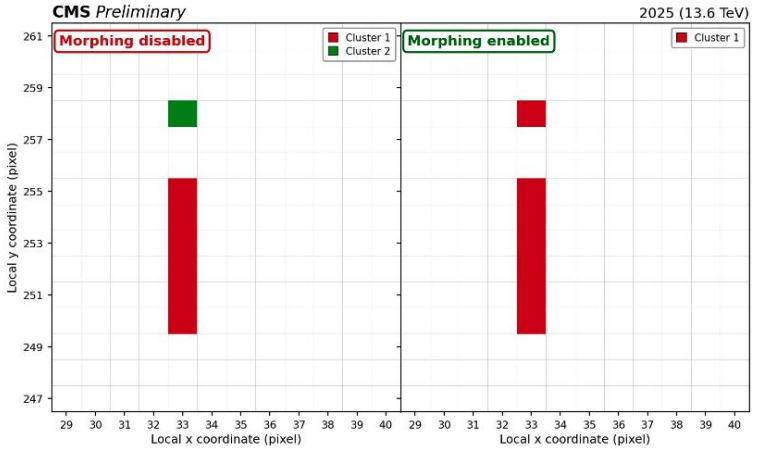


Problems with updating CPE conditions

- CPE conditions for start of 2025, based on end-of-2024 data, would not give improved tracking performance
  - Prompted a **very** long study in both pixel and alignment groups
- Several adaptations of the reconstruction to handle this
  - Updated template CPE centering procedure, turned off irradiation bias correction (IBC), deployed digi-morphing algorithm
- Some activities, e.g. producing CPE conditions, covered by a single expert
  - Large burden when extensive studies are needed but also risky for “standard” operations
  - **Avoid barely-covered tasks in Ph2**





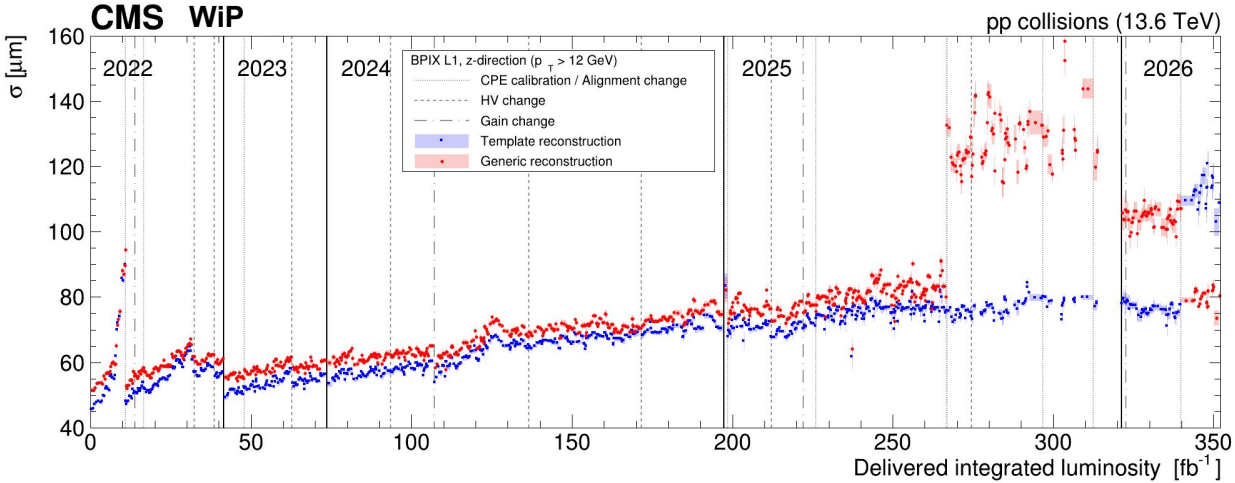


Henriette’s talk, Tracker Week Nov 2025

Digi-morphing DP note

Drifting CPE algorithms

- **Generic:** relies on cluster-end charge asymmetry and IBC for position prediction
  - Used in HLT and in track seeding for offline
- **Template:** fits cluster to precomputed cluster shapes based on simulated modelling
  - Used in offline, with track angle, for refining on-track hits
- With IBC off, generic ends up with angle-dependent bias w.r.t. the template algorithm
  - In offline, a single alignment is used in all tracking steps
- Turned off template reco in 2026 to have the same algorithm throughout the tracking chain
  - Alignment had absorbed the missing IBC; with template off, generic and alignment are consistent again
- The connection between alignment and pixel local reco needs to be kept in mind for Ph2 developments



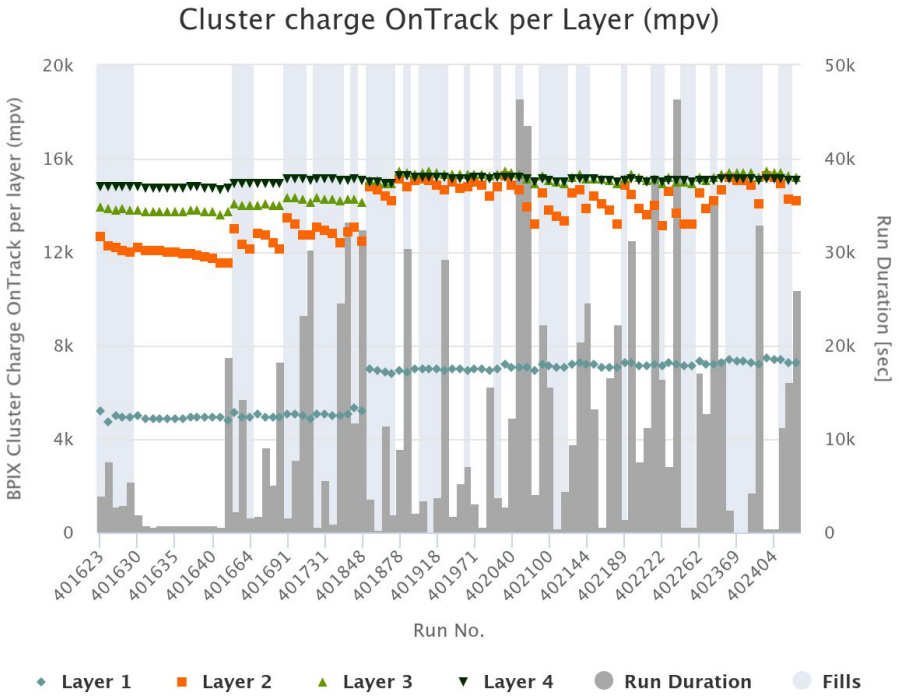
[2026 pixel performance](#) (not yet approved!)

Generic's resolution hurt by turning off IBC in mid-2025 since alignment prefers to tune to the final (template) algorithm

After disabling template CPE, alignment recovers the performance of Generic

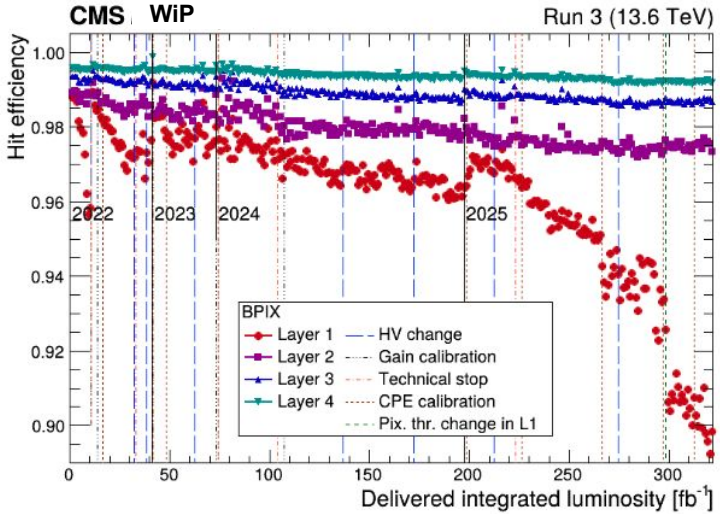
Suspicious gain calibration

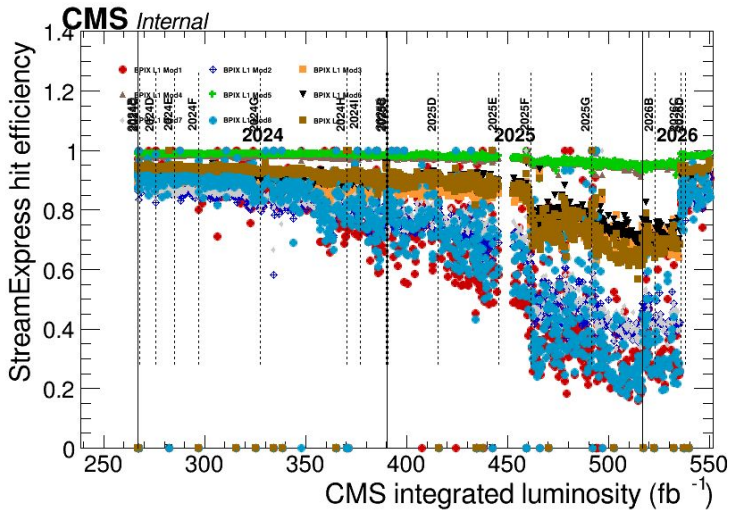
- In 2026, a gain calibration was performed, analyzed, deployed, and **reverted**
  - New payload was deployed but it resulted in a large charge drop
  - Operations thought there was an issue with the calibration itself
  - Later analysis showed that the gain calibration was correct
- The gain calibration was rarely done (1–2 times per year)
  - More frequent calibrations would prevent the large step which made us suspicious
    - Requires special calibration run so there are some complications
- Question to audience: I would like to know how gain calibration format should look like for Ph2 (please find me during a break)
  - Pedestal per pixel and gain per column would result in too large a payload



Disagreeing hit efficiency results

- Steady drop in hit efficiency seen in 2025, but on much different scales in DQM plots versus standard Pixel offline workflows
  - Took several months to understand the difference
    - Different requirements / assumptions that were no longer appropriate with heavy radiation damage
  - In the process, discovered that the calibration format (ALCARECO) lacks certain information (PR)
- Personal suggestion (no consensus in the offline group) is to centralize tools for monitoring and calibration
  - Pixel offline relies on privately maintained CMSSW-like packages (e.g. PixelNtuplizer)
- Root cause: lack of personpower delayed the investigation of the differences





Plots are not approved
Shown only to demonstrate the mismatch that we were seeing (different y scales!)

Personpower

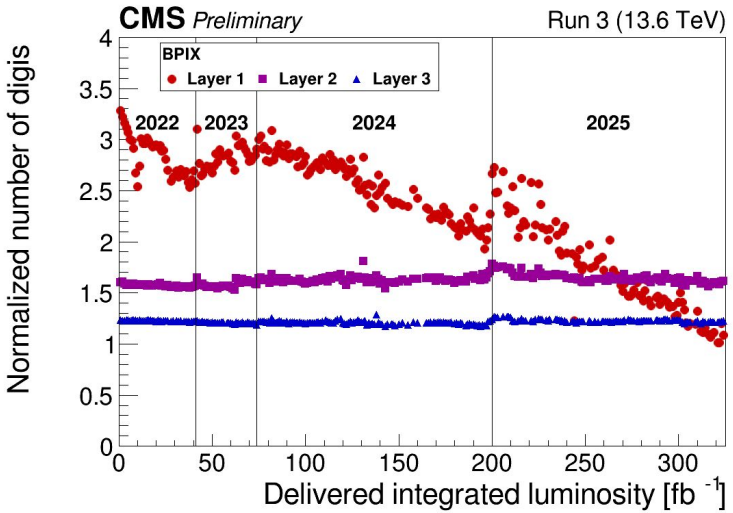
- Related to most (all?) of the points above
- Many tasks are covered by one person
- Same pair of conveners in entire Run 3
  - Healthy “rotation” of conveners would have ensured better diffusion of knowledge
  - Since end of 2025, only one convener (hopefully doing an adequate job)
    - **Luckily**, Dinko is still active within the Tracker DPG
- Not many newcomers which usually means conveners cover multiple tasks
- Algorithms and procedures written “before our time”
  - Indebted to Morris and Danek, guardians of historical pixel knowledge
- It is an excellent time to bring new people that are interested in preparing for Ph2 into the group

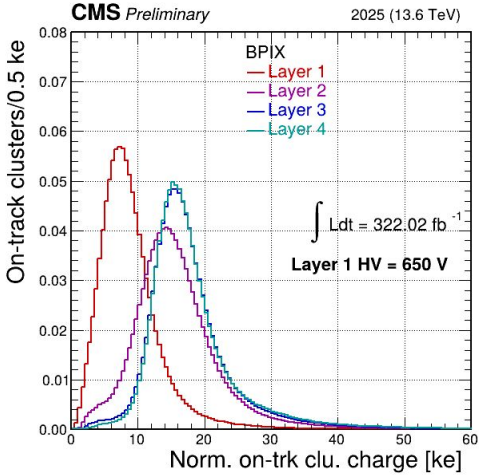


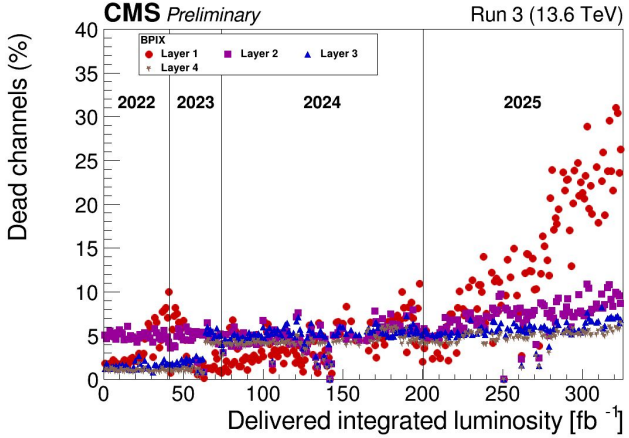


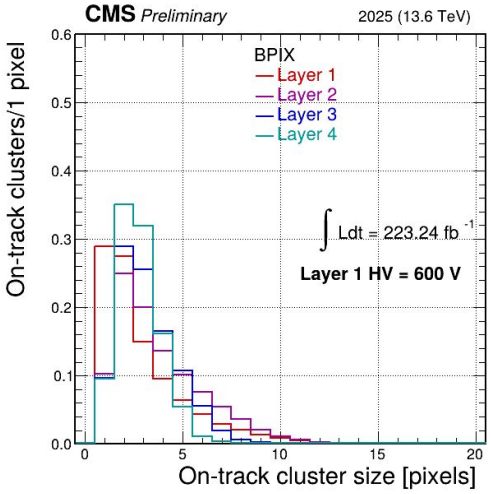
Extent of radiation damage

- Before concluding the talk, a reminder on what we had to fight against
  - Most of the problems are related to BPIX L1
- Understanding the damage we can expect in innermost Ph2 layer(s) should guide the preparation efforts



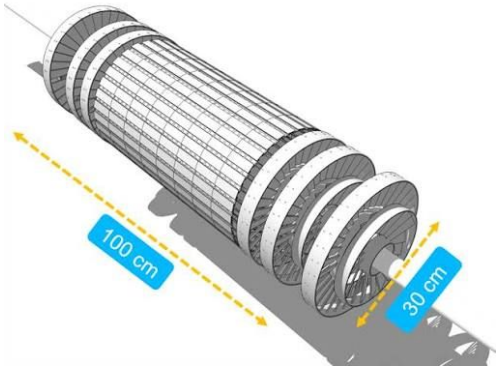




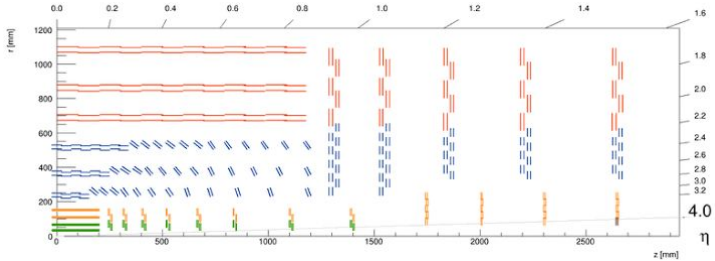


Summary

- Pixel detector delivered good data throughout Run 3, with L1 operating well beyond its design radiation tolerance
- Run 3 efforts do not stop with the end of collisions
  - Investigating the best conditions for and the potential gain of a ReReco of 2025 / 26 data
  - Preparing the 2026 performance note
  - Contributing to the tracking performance paper
- Some lessons for Ph2
  - Understanding the expected change due to radiation is key
    - The detector may change fast at the start, but we also need a plan for more severe radiation damage later on
  - Calibrations should be automated if possible
  - Local reconstruction and alignment are one system
    - To keep in mind when designing calibration procedures
  - Personpower: avoid single-expert tasks, rotate conveners, document procedures







Backup

What do we calibrate?

Fixed throughout Ph1
Probably the same for Ph2

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SiPixelLorentzAngleSim SiPixel2DTemplateDBObject (denominator)	MC
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SiPixelVCal	Data (outside GT)

Presented by Dinko at the Conditions preparations workshop (Indico)

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Analysis of a special gain calibration run
done once or twice per year
Ph2 readout might require different
payload format

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done once or twice per year
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payload format

Information on bad pixel components for
tracking to (not) expect valid hits
PCL automation with manual intervention
for known permanently dead modules
Automation to be used in Ph2 as well,
potentially expanded to HLT

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Modelling of efficiency dependence on the inst. luminosity. Derived once at beginning of Run 3
Not clear what form of this condition would be needed for Ph2

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"CPE conditions" or " templates"
Used for cluster parameter (position, uncertainty) estimation, and modelling radiation damage in simulation
Lengthy procedure with a lot of human oversight
Tested to work on Ph2 (more in talk by Morris today) but some degree of automation and centralization is desirable

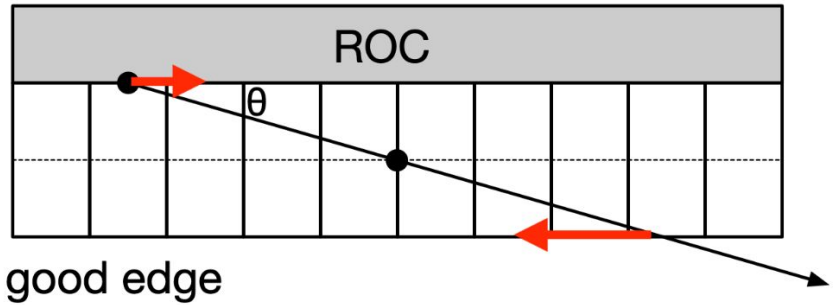
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Presented by Dinko at the Conditions preparations workshop (Indico)

CPE conditions workflow

1. **Data measurement:** obtain E-field vs depth from the collision data
2. **Sensor modelling:** a TCAD simulation of the irradiated sensor, tuned through its radiation-damage parameters
3. **Cluster simulation:** PixelAV simulates clusters using the model above. We compare them to clusters from data
    - Iterate steps 2–3 until data and simulation agree
4. **Template production:** simulate clusters over a grid of track angles and build the CPE templates
5. **Validation:** technical check that the templates work in CMSSW, measure the expected resolutions

Charge loss causes both edges to shift toward the cluster center



- The position reconstructed by the generic algorithm is dominated by the loss of charge [pixels] from the bad edge
  - shifts the reconstructed position to the left